Run report: itemF-control-run-20261018T2128Z

Descriptive record of one recorded run of examples/n2_routes.rs, mode control (Amendment F) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, 4 threads); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
commit c51acec659201b5aed5a412ca673ea3ccabfe761
cmd: n2 research/weil_descent_angles_20261018/results/itemF-control-run-20261018T2128Z control (N2_THREADS=4, sizes 20..40)
rustc 1.97.0 (2d8144b78 2026-07-07)
4
Sun Oct 18 21:28:51 UTC 2026
post-run (d2edb2ab8): the tool was rustfmt-formatted (format only; unit tests pass); outputs unaffected.
```

control_summary.txt

```
F bits 20 Jind: runs 9 steps/p mean 0.2949 sd 0.1192 | repeated divisors per run 0.11 | solved by a repeat 1/9
F bits 20 Jwalk: runs 9 steps/p mean 0.2439 sd 0.1300 | repeated divisors per run 0.33 | solved by a repeat 3/9
F bits 24 Jind: runs 9 steps/p mean 0.3221 sd 0.1617 | repeated divisors per run 0.11 | solved by a repeat 1/9
F bits 24 Jwalk: runs 9 steps/p mean 0.3651 sd 0.1595 | repeated divisors per run 0.33 | solved by a repeat 3/9
F bits 28 Jind: runs 9 steps/p mean 0.3920 sd 0.0727 | repeated divisors per run 0.00 | solved by a repeat 0/9
F bits 28 Jwalk: runs 9 steps/p mean 0.3434 sd 0.1207 | repeated divisors per run 0.22 | solved by a repeat 2/9
F bits 32 Jind: runs 9 steps/p mean 0.2897 sd 0.1118 | repeated divisors per run 0.33 | solved by a repeat 3/9
F bits 32 Jwalk: runs 9 steps/p mean 0.2999 sd 0.1097 | repeated divisors per run 0.67 | solved by a repeat 6/9
F bits 36 Jind: runs 9 steps/p mean 0.3703 sd 0.0986 | repeated divisors per run 0.22 | solved by a repeat 2/9
F bits 36 Jwalk: runs 9 steps/p mean 0.3608 sd 0.1270 | repeated divisors per run 0.33 | solved by a repeat 3/9
F bits 40 Jind: runs 9 steps/p mean 0.3174 sd 0.0639 | repeated divisors per run 0.11 | solved by a repeat 1/9
F bits 40 Jwalk: runs 9 steps/p mean 0.3300 sd 0.1517 | repeated divisors per run 0.44 | solved by a repeat 4/9
F by cofactor Jind h=4: runs 33 steps/p mean 0.3688 sd 0.0981 (s.e. 0.0171)
F by cofactor Jind h=8: runs 21 steps/p mean 0.2718 sd 0.1058 (s.e. 0.0231)
F by cofactor Jwalk h=4: runs 33 steps/p mean 0.3443 sd 0.1316 (s.e. 0.0229)
F by cofactor Jwalk h=8: runs 21 steps/p mean 0.2917 sd 0.1358 (s.e. 0.0296)
F all answers verified: true
```

control.csv (first 12 of 108 rows)

bits	instance	p	h	ell	target	arm	steps	relations	repeat_events	solved_by_repeat	verified
20	0	1091	8	148873	0	Jwalk	265	125	1	1	1
20	0	1091	8	148873	0	Jind	413	206	0	0	1
20	0	1091	8	148873	1	Jwalk	550	259	0	0	1
20	0	1091	8	148873	1	Jind	171	82	0	0	1
20	0	1091	8	148873	2	Jwalk	107	57	0	0	1
20	0	1091	8	148873	2	Jind	458	213	0	0	1
20	1	1061	8	140627	0	Jwalk	243	120	1	1	1
20	1	1061	8	140627	0	Jind	210	109	0	0	1
20	1	1061	8	140627	1	Jwalk	45	28	0	0	1
20	1	1061	8	140627	1	Jind	230	108	0	0	1
20	1	1061	8	140627	2	Jwalk	308	166	0	0	1
20	1	1061	8	140627	2	Jind	380	207	0	0	1

Graph

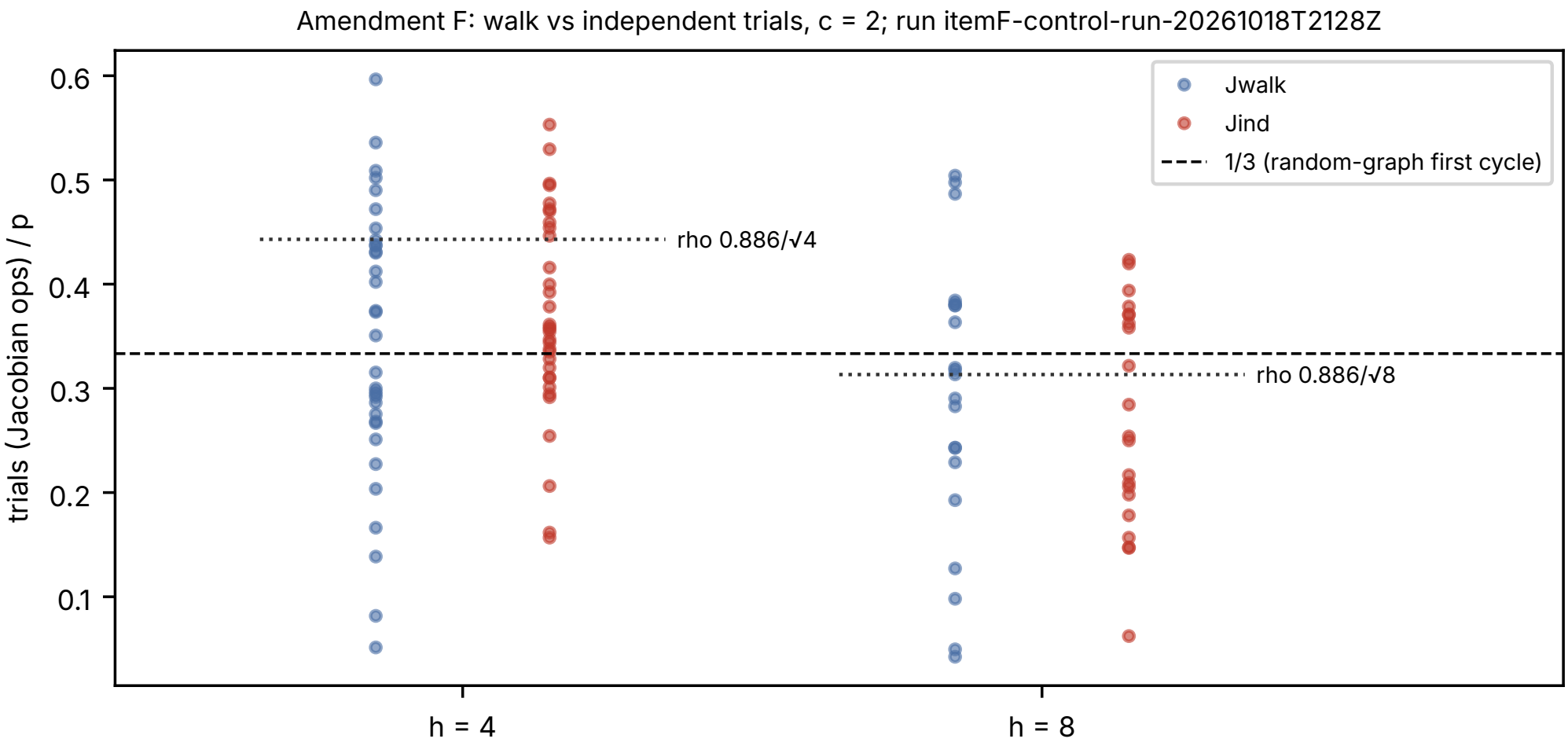


Figure 1: Amendment F: Arm J trials per p, walk vs independent trials, by cofactor h; each dot one verified run.